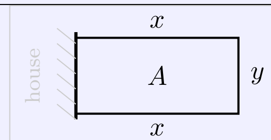
L2.6a Optimization

§3.7 — optimization equation, constraint, interval

The optimization template

	Step	**Sarah's fence** — 40 ft of fencing, the house is one side, and neither side may be under 5 ft
①	**Picture**	
②	**Sort the question**	**OEq** $A = xy$ **Constraint** $2x + y = 40$, so $y = 40 - 2x$ **Interval** $x \in [5, 17.5]$
③	**Find critical pts**	$A = x(40 - 2x) = 40x - 2x^2$ $A' = 40 - 4x = 0$, so $x = 10$ back into the constraint: $2(10) + y = 40$, so $y = 20$
④	**Does this Maximize/min?**	$A'' = -4$, so A is CD everywhere, so $x = 10$ is a Max *unsure which it is? that is what the 2nd Derivative Test is for*
⑤	**Check endpoints**	$A(5) = 150$ $A(10) = 200$ $A(17.5) = 87.5$

The question tells you what to optimize. The constraint tells you what to eliminate.

ELIMINATION WITH THE CONSTRAINT

An optimization equation in two variables cannot be differentiated. Use the constraint to write one variable in terms of the other, substitute, and *then* take the derivative.

$$\boxed{A = xy \quad \text{with} \quad y = 40 - 2x \quad \Longrightarrow \quad A(x) = 40x - 2x^2}$$

"if you find yourself differentiating the constraint, you have swapped the two"

THE INTERVAL SLOT

The interval is the third slot, and it is the one the picture does not always hand you. It comes from **both** sources at once — what the geometry allows, and what the question restricts.

from the geometry	a length cannot be negative, so $x \geqslant 0$
from the question	"at least 5 ft wide/long" gives $x \geqslant 5$ *and* $y \geqslant 5$

The second one binds at *both* ends. $x \geqslant 5$ reads off directly; and $y \geqslant 5$ with $2x + y = 40$ gives $2x + 5 \leqslant 40$, so $x \leqslant 17.5$.

$$\boxed{x \in [5,\ 17.5]}$$

"not every interval is $[0, \textit{something}]$ *— read the restrictions"*

ENDPOINT CHECK

A critical number is a candidate, not an answer. Evaluate the optimization equation at **every** critical number *and* at **both** ends, then compare the list.

$A(5) = 150$	$A(10) = \mathbf{200}$	$A(17.5) = \mathbf{87.5}$
left end	critical number	right end
	Max	**min**

The Max is interior, where the derivative put it. **The min is at the right endpoint** — and the derivative never proposed it. That is what step ⑤ is for.

Then **back-substitute**: $x = 10$ is not what was asked for. The question wanted the dimensions, so finish — $x = 10$ ft, $y = 20$ ft, largest area 200 ft^2 @ $x = 10$.

"an endpoint answer is not a mistake; it is often the answer"

WATCH OUT

✗ ~~the constraint is the thing you maximize~~ → **the OEq is what you optimize; the constraint is what lets you eliminate**

✗ ~~$A = xy$, so $A' = x'y + xy'$~~ → **eliminate first. One variable, then differentiate — nothing here is a function of t**

✗ ~~the critical number is the answer~~ → **it is a candidate. Step ④ or step ⑤ still has to say which kind it is**

✗ ~~the interval is always $[0, \text{something}]$~~ → **read the restrictions. Sarah's is $[5, 17.5]$, and the min lives at the right end**

✗ ~~an open interval means you skip the justification~~ → **it means there are no ends to check, so the justification is the whole argument — use the 2nd Derivative Test**

✗ ~~x is the answer~~ → **back-substitute. The question asked for the dimensions, the cost, or the time**